Name Owen Hutflesz	**Version** ● B C D E
ID	**Other**
Section	**Marking Instructions** Be sure to completely fill in the appropriate bubble. **Example** ● B C D E
Date	

	A	B	C	D	E
1	A	B	C	●	E
2	A	B	●	D	E
3	A	B	C	●	E
4	A	●	C	D	E
5	●	B	C	D	E
6	A	●	C	D	E
7	●	B	C	D	E
8	●	B	C	D	E
9	●	B	C	D	E
10	●	B	C	D	E
11	A	●	C	D	E
12	A	●	C	D	E
13	●	B	C	D	E
14	A	●	C	D	E
15	●	B	C	D	E
16	●	B	C	D	E
17	A	●	C	D	E
18	A	●	C	D	E
19	A	●	C	D	E
20	A	B	C	D	E
21	A	B	C	D	E
22	A	B	C	D	E
23	A	B	C	D	E
24	A	B	C	D	E
25	A	B	C	D	E

	A	B	C	D	E
26	A	B	C	D	E
27	A	B	C	D	E
28	A	B	C	D	E
29	A	B	C	D	E
30	A	B	C	D	E
31	A	B	C	D	E
32	A	B	C	D	E
33	A	B	C	D	E
34	A	B	C	D	E
35	A	B	C	D	E
36	A	B	C	D	E
37	A	B	C	D	E
38	A	B	C	D	E
39	A	B	C	D	E
40	A	B	C	D	E
41	A	B	C	D	E
42	A	B	C	D	E
43	A	B	C	D	E
44	A	B	C	D	E
45	A	B	C	D	E
46	A	B	C	D	E
47	A	B	C	D	E
48	A	B	C	D	E
49	A	B	C	D	E
50	A	B	C	D	E

	A	B	C	D	E
51	A	B	C	D	E
52	A	B	C	D	E
53	A	B	C	D	E
54	A	B	C	D	E
55	A	B	C	D	E
56	A	B	C	D	E
57	A	B	C	D	E
58	A	B	C	D	E
59	A	B	C	D	E
60	A	B	C	D	E
61	A	B	C	D	E
62	A	B	C	D	E
63	A	B	C	D	E
64	A	B	C	D	E
65	A	B	C	D	E
66	A	B	C	D	E
67	A	B	C	D	E
68	A	B	C	D	E
69	A	B	C	D	E
70	A	B	C	D	E
71	A	B	C	D	E
72	A	B	C	D	E
73	A	B	C	D	E
74	A	B	C	D	E
75	A	B	C	D	E

	A	B	C	D	E
76	A	B	C	D	E
77	A	B	C	D	E
78	A	B	C	D	E
79	A	B	C	D	E
80	A	B	C	D	E
81	A	B	C	D	E
82	A	B	C	D	E
83	A	B	C	D	E
84	A	B	C	D	E
85	A	B	C	D	E
86	A	B	C	D	E
87	A	B	C	D	E
88	A	B	C	D	E
89	A	B	C	D	E
90	A	B	C	D	E
91	A	B	C	D	E
92	A	B	C	D	E
93	A	B	C	D	E
94	A	B	C	D	E
95	A	B	C	D	E
96	A	B	C	D	E
97	A	B	C	D	E
98	A	B	C	D	E
99	A	B	C	D	E
100	A	B	C	D	E

NAME: Student ID:

CSE 16-1 Spring 2026 Midterm # 1

April 28th, 2026

Topic	Points	Section	Points
Part A: Nested Quantifiers	6	Part B: Set & Logic	16
Part C: Counting	20	Part D:Truth Table	10
		Total	**52**

Put the version of the exam on top of the bubble sheet. For questions with multiple choice answers, please fill in the bubble on the bubble sheet with a dark colored pen.

Part A: Nested Quantifiers

[6 points]

1. For this English statement, find the mathematical statement that corresponds to it: for any set A in the universe U and any set B in the universe U, the cardinality of the power set of the union of A and B is 2 to the power of the cardinality of A plus the cardinality of B minus the cardinality of the intersection of A and B.

   (A) $\forall A \subseteq U, \forall B \subseteq U, |\mathcal{P}(A \cap B)| = 2^{|A|+|B|-|A \cup B|}$

   (B) $\exists A \subseteq U, \exists B \subseteq U, |\mathcal{P}(A \cup B)| = 2^{|A|+|B|-|A \cap B|}$

   (C) $\forall A \subseteq U, \forall B \subseteq U, |A \cup B| = 2^{|A|+|B|-|A \cap B|}$

   (D) $\forall A \subseteq U, \forall B \subseteq U, |\mathcal{P}(A \cup B)| = 2^{|A|+|B|-|A \cap B|}$ D

   (E) None of the above

Q1-D: "for any A,B in U" = ·A,·B (not ·). |P(A·B)| = 2^|A·B| = 2^(|A|+|B|-|A·B|). Only (D) has both · · quantifiers AND the power-set P( ) wrapper. ·

2. For natural numbers n and k, we are given the set $A(n,k)$ of non-decreasing sequences of n natural numbers $a_1 \leq a_2 \leq \ldots \leq a_n$ such that $a_n = k$. What is the negation of the statement:

   $\forall n \in \mathbb{N}$ with $n > 1, \forall k \in \{n, n+1, \ldots, 2n\}, \forall (a_1, a_2, \ldots, a_n) \in A(n,k), \exists i \in \{1, 2, ..., n-1\}, a_i = a_{i+1}$

   (A) $\exists n \in \mathbb{N}$ with $n > 1, \exists k \in \{n, n+1, \ldots, 2n\}, \exists (a_1, a_2, \ldots, a_n) \in A(n,k), \exists i \in \{1, 2, ..., n-1\}, a_i \neq a_{i+1}$

   (B) $\exists n \in \mathbb{N}$ with $n > 1, \forall k \in \{n, n+1, \ldots, 2n\}, \exists (a_1, a_2, \ldots, a_n) \in A(n,k), \forall i \in \{1, 2, ..., n-1\}, a_i = a_{i+1}$

   (C) $\exists n \in \mathbb{N}$ with $n > 1, \exists k \in \{n, n+1, \ldots, 2n\}, \exists (a_1, a_2, \ldots, a_n) \in A(n,k), \forall i \in \{1, 2, ..., n-1\}, a_i \neq a_{i+1}$ C

   (D) $\exists n \in \mathbb{N}$ with $n > 1, \exists k \in \{n, n+1, \ldots, 2n\}, \forall (a_1, a_2, \ldots, a_n) \in A(n,k), \forall i \in \{1, 2, ..., n-1\}, a_i \neq a_{i+1}$

   (E) None of the above

Q2-C: Negate ·n,·k,·seq,·i (a·=a···). Each ···, each ···, negate predicate: ·n,·k,·seq,·i (a··a···). That is exactly (C). ·

3. For this English statement, find the mathematical statement that corresponds to it: For all real numbers y, there exists another real number x, for which there exists an integer z, such that y is the product of z by x squared.

   (A) $\forall y \in \mathbb{R}, \exists\, x \in \mathbb{R}, \exists z \in \mathbb{Z}, x = zy^2$

   (B) $\forall y \in \mathbb{R}, \exists\, x \in \mathbb{R}, \exists z \in \mathbb{Z}, y \neq zx^2$

   (C) $\forall y \in \mathbb{R}, \forall\, x \in \mathbb{R}, \exists z \in \mathbb{Z}, y = zx^2$

   (D) $\forall y \in \mathbb{R}, \exists\, x \in \mathbb{R}, \exists z \in \mathbb{Z}, y = zx^2$ D

   (E) None of the above

Q3-D: "For all y" = ·y··, "there exists x" = ·x··, "there exists integer z" = ·z···. Predicate: y = z·x². That gives ·y··, ·x··, ·z··, y=zx² · (D). ·

Part B: Sets & Logic

Please check all statements below and check (A) for True and (B) for False.
U is the universe that contains all possible sets, and $\bar{X}$ is the complement of the set X in the universe U.**[16 points]**

(B) 4. $\forall A \subset U,\ A \setminus A^C = \emptyset$ — A\A· = A·(A·)· = A·A = A (not ·)

(A) 5. The biconditional operator $\Leftrightarrow$ is associative. — · is associative ·

(B) 6. $|\mathcal{P}(\{1, \{1, 2\}, \{1\}\})| = 2$ — Set has ·3 distinct elements · |P|·4 · 2

(A) 7. $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$ — Standard P(A·B)=P(A)·P(B) identity ·

(A) 8. $\{n^2 + 3 : n \in \mathbb{N}\} = \{z^2 + 2z + 4 : z \in \mathbb{Z} \text{ and } z \geq 0\}$ — z²+2z+4=(z+1)²+3, let n=z+1 · same set ·

(A) 9. $(P \Rightarrow Q)$ is equivalent to $(\neg P \vee Q)$ — P·Q · ¬P·Q ·

(A) 10. If P is true, then $P \vee Q$ is true — If P=T then P·Q=T for any Q ·

(B) 11. $\neg(P \wedge \neg Q)$ is equivalent to $(\neg P \wedge Q)$ — ¬(P·¬Q) = ¬P·Q (not ¬P·Q)

(B) 12. $\emptyset \in \{\{0\}, \{1\}, \{\emptyset\}\}$ — 0·{{0},{1},{·}} · elements are SETS {0},{1},{·}; none equals 0

(A) 13. $|\mathcal{P}(A \cup B)| = \frac{2^{|A|}2^{|B|}}{2^{|A \cap B|}}$ — |P(A·B)|=2^(|A|+|B|-|A·B|)=2^|A|·2^|B|/2^|A·B| ·

(B) 14. $(A \subseteq B) \Rightarrow (A \setminus B = A)$ — A·B · A\B=· (not A)

(A) 15. $(A \subseteq B) \Rightarrow (A \cap B = A)$ — A·B · A·B=A ·

(A) 16. If $A = \{1, 2, 3\}$ and $B = \{2, 3, 1\}$, then $A = B$ — Sets are unordered · {1,2,3}={2,3,1} ·

(B) 17. If $B = C \cap D, A \times B = (A \times C) \cup (A \times D)$ — A×(C·D)=(A×C)·(A×D); exam uses · · FALSE

(B) 18. $|\{\emptyset\}| = 0$ — |{·}|=1, not 0 (one element: the empty set)

(B) 19. Let $A = \{2, \{2, \{2\}\}\}$ and $B = \{2, \{2\}\}$. Is $B \subseteq A$? — {2}·A={2,{2,{2}}} · B·A

Part C: Counting

The Cruz-Hack Club organized a hackathon. Four teams participated (Cruzers, K-Combo, CodeCode, Sluggies), for a total of 11 student participants (that is a set of hackers $H = \{h_1, h_2, \ldots, h_{11}\}$ and five judges (that is, a set $J = \{j_1, j_2, \ldots, j_5\}$) were invited to rate the projects. When voting, votes are anonymous.

20. Assuming all the students and the judges enter the Hackathon room one by one, how many ways are there for them to enter in such a way that no two judges follow each other (that is, in such a way that there is at least one student in between the judges). **[2 points]**

ANSWER: **11! × C(12,5) × 5!**

Method (Stars & Bars + arrangement):
Step 1: Seat 11 students in a row · 11! ways.
Step 2: 11 students create 12 gaps (_S_S_·_S_).
Step 3: Choose 5 of the 12 gaps for judges · C(12,5).
Step 4: Arrange 5 judges in the chosen gaps · 5!.
Total = 11! × C(12,5) × 5! [guarantees no two judges adjacent]

21. In the room, there are 16 tables for the students to sit at. Each table sits 2 people. How many students would we need to ensure that 2 students sit at the same table? [**2 points**]

ANSWER: **17**

Pigeonhole Principle:
16 tables × max 1 student/table = 16 students, no table has 2. (worst case)
One more student (17th) forces some table to have ·2. · Answer: 16+1 = 17

22. For some awards, all participants and judges (for a total of 16 votes) get to vote. How many ways are there to split the vote among the four projects, considering it is possible a project receives no votes? [**2 points**]

ANSWER: **C(19, 3) = 969**

Stars & Bars, teams may receive 0 votes:
n=16 votes into k=4 teams (0 allowed) · C(n+k-1, k-1) = C(16+3, 3) = C(19,3)
C(19,3) = 19·18·17 / 6 = 969

23. One of the project of the Hackathon answers this question: given k distinct natural numbers $a_1, a_2, \ldots, a_k$ between 1 and 32, what is the minimum value of k such that there are two consecutive numbers in the set $\{a_1, a_2, \ldots, a_k\}$? [**2 points**]

ANSWER: **17**

Max subset of {1..32} with no two consecutive = all odd numbers {1,3,5,·,31} = 16 nums.
Adding any 17th number creates a consecutive pair (every even is adjacent to an odd).
Minimum k = 16 + 1 = 17 (Pigeonhole: 16 pairs, 17 elements forces a pair)

24. Assuming 16 people will vote for any of the 4 teams again, how many votes do we need to count to guarantee that at least one team has received three votes? [**2 points**]

ANSWER: **9**

Pigeonhole: Give each of 4 teams exactly 2 votes = 8 total. No team has ·3. (worst case)
One more (9th vote) forces some team to ·3. · Answer: 2×4 + 1 = 9

25. Assuming the 16 voters can vote for any team, what can you say about the minimum number of votes to count such that all team receive at least one vote? [**2 points**]

ANSWER: **Cannot be guaranteed.**

Voters choose freely · all 16 votes could go to one team (distribution: 16,0,0,0).
No constraint forces any team to receive a vote, no matter how many votes are counted.
· It is IMPOSSIBLE to guarantee every team gets ·1 vote.

26. The organizers do not want a team to have no vote. They collude with the judges in order to achieve this outcome. How many ways are there to split the 16 votes such that each of the 4 teams has at least one vote? [**2 points**]

ANSWER: **C(15, 3) = 455**

Stars & Bars, each team gets ·1 vote:
Pre-assign 1 vote to each team (4 votes used, 12 remain). Distribute 12 freely among 4.
Equivalently: C(V-1, k-1) = C(16-1, 4-1) = C(15,3)
C(15,3) = 15·14·13 / 6 = 455

27. For some other award, the organizers do not want all the participants to vote, but only the judges and 5 students. How many ways are there to select the 5 voting students among all 11 students? **[2 points]**

ANSWER: **C(11, 5) = 462**

Choose 5 voting students from 11 total students.
Unordered selection, no repetition · C(n,r) = C(11,5)
C(11,5) = 11·10·9·8·7 / (5·4·3·2·1) = 462

28. For one of the awards, the 16 votes are tallied and read in order. How many possible lists of votes can be read in such manner, considering each vote is one of the four project names, and that the same project can be voted for repeatedly.

ANSWER: **4^16**

Ordered list of 16 votes; each vote independently chooses 1 of 4 teams.
Ordered sequence with repetition: n^r = 4^16
4^16 = 4,294,967,296

29. The organizers will select three participants to represent the Cruz-Hack Club at the state level. They will select 3 students from both the winning team and the runner-up team. Both these teams have four members each. The organizers ensure that at least one of the selected students is from the winning team, and the other two selected students can come from either team. How many ways are there to select these 3 students? **[2 points]**

ANSWER: **52**

Complement method: Pool = {4 winning team members} + {4 runner-up members} = 8 students.
Total ways to choose 3 from 8: C(8,3) = 56
Ways with 0 from winning team (all from runner-up): C(4,3) = 4
At least 1 from winning team: 56 · 4 = 52

Part D: Truth Table

Write the truth table to answer the following question: Is $(P \wedge (Q \Leftrightarrow R)) \Rightarrow (\neg P \Rightarrow (Q \vee R))$ a tautology, that is a proposition that is always true. The truth table should have columns for (1): $(Q \Leftrightarrow R)$, (2): $(Q \vee R)$, (3): $(P \wedge (Q \Leftrightarrow R))$, (4): $(\neg P \Rightarrow (Q \vee R))$ and (5): $(P \wedge (Q \Leftrightarrow R)) \Rightarrow (\neg P \Rightarrow (Q \vee R))$ (2 points per column). Based on the truth table, state your answer to the question. [**10 points**]

P	Q	R	(1) Q<=>R	(2) QvR	(3) P^(Q<=>R)	(4) ~P=>(QvR)	(5) (P^(Q<=>R)) =>(~P=>(QvR))
T	T	T	T	T	T	T	T
T	T	F	F	T	F	T	T
T	F	T	F	T	F	T	T
T	F	F	T	F	T	T	T
F	T	T	T	T	F	T	T
F	T	F	F	T	F	T	T
F	F	T	F	T	F	T	T
F	F	F	T	F	F	F	T

Conclusion: Column (5) is TRUE in all 8 rows.
Therefore: (P ^ (Q<=>R)) => (~P => (Q v R)) is a TAUTOLOGY.